Project 3 Harmonic Analysis

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 < p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p \, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p \, w(x)\,dx\right)^{1/p}$.

The central object of this report is the Hardy–Littlewood maximal operator M, which assigns to each point the largest average of $|f|$ over cubes containing that point. In the proof of our main result we will also need the analogous operator in which these averages are taken with respect to the measure $w(x)\,dx$ instead of Lebesgue measure.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy, \tag{1}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. If the supremum is taken only over cubes Q centred at x, we write M^c_w. If $w \equiv 1$ we drop the subscript and write M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) on $L^p(\mathbb{R}^d)$ for $1 < p \leq \infty$ [2]. In the light of our $L^p_w(\mathbb{R}^d)$ definition a natural question arises, for fixed $1 < p < \infty$, for which weights w is the sublinear operator M bounded on $L^p_w(\mathbb{R}^d)$? In other words, for which w is there a constant C such that for all $f \in L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w\,dx \right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p \, w\,dx \right)^{1/p} = C\,\|f\|_{L^p_w}. \tag{2}$$

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$, with the essential supremum taken with respect to $w(x)\,dx$, the answer is trivial: M is bounded if and only if $w > 0$ almost everywhere. If $w > 0$ a.e., then $w(x)\,dx$ and dx have the same null sets, so $\|Mf\|_{L^\infty} \leq \|f\|_{L^\infty}$ applies. If $w = 0$ on a set Z with $|Z| > 0$, then $f = \chi_Z$ has weighted supremum 0, while $Mf > 0$ everywhere. Hence we restrict to $1 < p < \infty$.*

Suppose that (2) holds for a weight w. A first observation is that w cannot vanish on a set of positive measure. The indicator function of a bounded part of such a set has norm zero in L^p_w, while its maximal function is positive everywhere, which contradicts (2). More generally, w cannot be much smaller on part of a cube than on the rest of it. A function concentrated where w is small has a small norm in L^p_w, while its maximal function is large on the whole cube, where w is large. Making this precise leads to the following necessary condition.

Proposition 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (2) holds with constant C, then for every cube Q we have $\int_Q w^{-\frac{1}{p-1}}\,dx < \infty$ and*

$$\left(\frac{1}{|Q|}\int_Q w\,dx\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dx\right)^{p-1} \leq C^p. \tag{3}$$

In particular, $w > 0$ almost everywhere.

Thus, if (2) holds for a weight w, then by Proposition 1, w satisfies (3) for every cube Q, with the same constant C.This gives rise to the next definition by Muckenhoupt [4].

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{4}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (2) holds with constant C, then $w \in A_p$ and*

$$[w]_{A_p} \leq C^p. \tag{5}$$

Corollary 1 shows that $w \in A_p$ is a necessary condition for (2) to hold. It is natural to ask whether it is also sufficient, that is, whether for every $w \in A_p$ there is a constant C such that (2) holds. Since Corollary 1 moreover shows that any such C must satisfy $C \geq [w]^{1/p}_{A_p}$, we may ask more precisely how large C has to be in terms of $[w]_{A_p}$. The following theorem, due to Buckley [1], answers both questions.

Theorem 1 (Buckley; Theorem C in [3])**.** *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \leq c_{p,d}\,[w]^{\frac{1}{p-1}}_{A_p}\,\|f\|_{L^p_w}.$$

Moreover, the exponent $\frac{1}{p-1}$ is best possible: for every $\alpha < \frac{1}{p-1}$ there is no constant c such that $\|Mf\|_{L^p_w} \leq c\,[w]^{\alpha}_{A_p}\|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.

In particular, Theorem 1 shows that $w \in A_p$ is also a sufficient condition, since if $[w]_{A_p} < \infty$, then (2) holds with $C = c_{p,d}[w]_{A_p}^{1/(p-1)}$. Together with Corollary 1, this answers the question posed at the beginning completely and implies the following result.

Theorem 2 (Muckenhoupt)**.** *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem 2 is the original result of [4], published in 1972. His proof, as well as Buckley's proof of Theorem 1, relies on the fact that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$, which is a consequence of the reverse Hölder inequality for A_p weights. Buckley's proof moreover uses interpolation. We present neither of these proofs. For the upper bound in Theorem 1 we instead present the elementary proof of Lerner [3], which avoids the $A_{p-\varepsilon}$ property, and as noted above Theorem 2 then needs no separate proof since it follows immidiatly from Corollary 1 and Theorem 1. We complement Lerner's argument with a proof of Proposition 1 and an example showing that the exponent in Theorem 1 cannot be improved.

2 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$, let $y \in Q$, and let Q_y be the cube centred at y with side length $2\ell(Q)$. Then $Q \subseteq Q_y \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z_i - y_i| \leq \ell(Q)$ for $i = 1, \dots, d$, since both z and y lie in Q. Hence $Q \subseteq Q_y$. Let c be the centre of Q. Then $|y_i - c_i| \leq \frac{1}{2}\ell(Q)$ for $i = 1, \dots, d$, so every $z \in Q_y$ satisfies

$$|z_i - c_i| \leq |z_i - y_i| + |y_i - c_i| \leq \ell(Q) + \tfrac{1}{2}\ell(Q) = \tfrac{3}{2}\ell(Q)$$

for $i = 1, \dots, d$, that is, $z \in 3Q$. □

Lemma 2. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \leq 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, let Q_x be the cube centred at x with side length $2\ell(Q)$. By Lemma 1, $Q \subseteq Q_x$, and since $|Q_x| = 2^d|Q|$ we obtain

$$Mf(x) = \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\, dy \leq \sup_{Q \ni x} \frac{2^d}{|Q_x|} \int_{Q_x} |f(y)|\, dy \leq 2^d M^c f(x),$$

where the last inequality holds because each Q_x is a cube centred at x. □

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere, and $\sigma := w^{-\frac{1}{p-1}}$ is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q.*

Proof. Suppose that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, there is also a set F with $|F| > 0$ on which $w > 0$. Choose bounded subsets $E' \subseteq E$ and $F' \subseteq F$ of positive measure, and a cube Q containing both. Then $w(Q) \geq w(F') > 0$, while $\sigma = \infty$ on E', so $\sigma(Q) = \infty$. Hence the quantity in (4) is infinite for this cube, which contradicts $w \in A_p$. Thus $w > 0$ almost everywhere, and in particular $w(Q) > 0$

for every cube Q. Since w is locally integrable, $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. Finally, by the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty$$

for every cube Q. Since every compact set is contained in a cube, σ is locally integrable. □

References

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[3] Andrei K. Lerner, *An elementary approach to several results on the Hardy–Littlewood maximal operator*, Proc. Amer. Math. Soc. **136** (2008), no. 8, 2829–2833.

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